STANLEY NICHOLSON

stanley_nicholson@brown.edu ◇ stanleynicholson.com

SUMMARY

Applied mathematics Ph.D. candidate building mechanistic, data-driven models of cell differentiation and embryonic development.

EDUCATION

Brown University *Aug 2023 – May 2028 (expected)*

Ph.D. Candidate, Applied Mathematics; NSF Graduate Research Fellow GPA: 4.0/4.0

Illinois Institute of Technology *Aug 2020 – May 2023*

Joint B.S./M.S., Applied Mathematics; Minors in Computer Science, Bioinformatics GPA: 3.96/4.0

RESEARCH EXPERIENCE

Graduate Researcher, Brown University *Aug 2023 – Present*

Differentiation dynamics from single-cell RNA-seq Advisor: Björn Sandstede

- Inferring differentiation dynamics and topology (branch points, fate decisions) in hematopoiesis from scRNA-seq data.
- Independently formulated the project, combining optimal transport with differential equations.

Summer Research Intern, Flatiron Institute (Center for Computational Biology) *May 2025 – Present*

Early embryogenesis of the fruit fly PI: Stanislav Shvartsman

- Analyze light-sheet imaging data to identify biophysical mechanisms.
- Develop a hybrid differential-equation model of cytoskeletal and cell-cycle dynamics.
- Designed optogenetic experiments to validate the model.
- Collaborating with two Ph.D. candidates and a postdoctoral fellow.

PUBLICATIONS

- Peter Sentz, **Stanley Nicholson**, Yujin Cho, et al. "Learning Thermodynamic Master Equations for Open Quantum Systems". *Quantum* (2026).
- David A. Cooper, Joseph DePaolo-Boisvert, **Stanley A. Nicholson**, et al. "Intracellular Pocket Conformations Determine Signaling Efficacy through the μ Opioid Receptor". *Journal of Chemical Information and Modeling* (2025).
- Van Ngoc Thuy La, **Stanley Nicholson**, Amna Haneef, et al. "Inclusion of Control Data in Fits to Concentration–Response Curves Improves Estimates of Half-Maximal Concentrations". *Journal of Medicinal Chemistry* (2023).
- **Stanley Nicholson**, David D. L. Minh, and Robert Eisenberg. "H-Bonds in Crambin: Coherence in an α-Helix". *ACS Omega* (2023).

TALKS, MENTORING, AND TEACHING

Talk: Machine Learning of Cytoskeletal Machines (Cell Migration and Mitosis), NITMB *Nov 2025*

Mentor: directed readings (10 students) in computational biology, dynamical systems, and optimal transport

Teaching Assistant: Statistical Inference II; Computational Probability and Statistics

HONORS AND ENTREPRENEURSHIP

- Camras Scholar (full-tuition merit scholarship), Illinois Institute of Technology
- Co-founder, Receptify, supporting survivors of campus sexual violence; enabled by funding via pitch competitions